

Seiberg-Witten Theory

Ikhan Choi
Lectured by Konno Hokuto
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1 Day 1: April 7

cancelled: 4/28, 5/19

References

- Morgan “The Seiberg-Witten equations and applications to the topology of smooth 4-manifolds”
- Sasahira “Saibaagu-wittenhouteishiki” (Japanese)
- Matsumoto “Yojigentayoutai I,II”

Plan

- Introduction/results from SW theory
- SW equations
- SW invariant (definition/properties)
- Finite-dimensional approximation of the SW equation
- Bauer-Furuta invariant, Donaldson’s diagonalization theorem

Introduction

SW theory $\subset$ Gauge theory. We do not regard physical aspects.

Gauge theory: Using some non-linear PDE, study the topology of smooth 4-manifolds.

Two big methods:

1. By “counting” solutions to the PDE, construct an invariant of smooth 4-manifolds. Distinguish smooth 4-manifolds (detect exotic 4-manifolds, and two smooth manifolds are called *exotic* if they are homeomorphic but not diffeomorphic)
2. By studying the “solution space”(moduli space) for the PDE, give a constraint on the topology of smooth 4-manifolds. (Donaldson) Detect non-smoothable topological 4-manifolds, such manifolds are known to exist for dimension ≥ 4 .

More about 1. There is a fact that a compact topological manifold of dimension $\neq 4$ admits at most finitely many smooth structures. (cf. Kirby-Siebenmann theory in $\dim > 4$.) It is still difficult to compute the number in general. On the other hand, there are many compact 4-manifolds that admit infinitely many smooth structures. It is shown by gauge theoretic invariants such as Donaldson, SW, Bauer-Furuta. For example, Kodaira showed the K3 surface is unique up to diffeomorphism (of course uncountably many up to isomorphism). If $K3 := \{[z_0 : \cdots : z_3] \in \mathbb{CP}^3 : z_0^4 + \cdots + z_3^4 = 0\}$, then it is known to admit infinitely many smooth structures. (cf. log transform, (elliptic fibration $\rightarrow$ new elliptic fibration, this is closed in complex geometry, and Kronheimer showed $SO(3)$ -Donaldson invariant), knot-surgery)

More about 2. Donaldson’s diagonalization theorem and Furuta 10/8 inequality.

It is an open problem that asks if every smoothable topological 4-manifold has infinitely many smooth structures. Bauer-Furuta invariant gives a hint, but it is negative to be solved in this way... In experience true, knot theoretically? Gauge theory is effective when the intersection form is far from trivial. Example that we do not know if it is exotic: S^4 , $\mathbb{CP}^2$, $S^2 \times S^2$, $\mathbb{CP}^2 \# \overline{\mathbb{CP}}^2$.

1.1 Intersection form

Let X be an oriented closed topological 4-manifold. Then, $Q_X : (H^2(X, \mathbb{Z})/\text{Tor})^{\otimes 2} \rightarrow \mathbb{Z} : \alpha \otimes \beta \mapsto \langle \alpha \smile \beta, [X] \rangle$ is called the *intersection form*, and it is clearly homotopy invariant. As a remark, by the Poincaré duality we can identify $H^2(X, \mathbb{Z})$ with $H_2(X, \mathbb{Z})$, and $Q_X(\alpha, \beta)$ means $\#(\Sigma \cap \Sigma')$ counting with sign, where $\alpha = [\Sigma]$ and $\beta = [\Sigma']$ with Σ and Σ' oriented closed embedded surfaces in X (existence of such representative may not be clear for topological cases, but relatively clear for smooth cases.).

For example if $X = S^4$, then $H^2(S^4) = 0$ and $Q_{S^4} = *$ with size zero.

For example if $X = \mathbb{CP}^2$, then $H_2(\mathbb{CP}^2) = \mathbb{Z}[\mathbb{CP}^1]$ and $Q_{\mathbb{CP}^2} = (1)$ with size one.

For example if $X = S^2 \times S^2$, then $H_2(S^2 \times S^2) = \mathbb{Z}[S^2 \times *] \oplus \mathbb{Z}[* \times S^2]$ and $Q_{S^2 \times S^2} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. We can check self-intersection vanishes and off-diagonal part has a single intersection. (perturbing using trivial normal bundle?)

In general, Q_X is symmetric. If we consider the same device for 2-dimensional, it becomes anti-symmetric. It is also bilinear over $\mathbb{Z}$ and non-degenerate.

Exercise 1.1. The intersection form is non-degenerate. Use the Poincaré duality.

There is a converse question. When is a non-degenerate symmetric bilinear form over $\mathbb{Z}$ the intersection form of a topological 4-manifold?

Theorem 1.2 (Freedman, 1982). *Let Q be a non-degenerate symmetric bilinear form over $\mathbb{Z}$. Then, there is a topological simply connected closed orientable 4-manifold X such that $Q_X \cong Q$ over $\mathbb{Z}$. Moreover, such X are at most two up to homeomorphism, and determined by the Kirby-Siebenmann invariant $KS(X) \in H^4(X, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$.*

(proved by wild topology such as approximation with Cantor sets, and there would be almost no people understanding the proof.)

This theorem says that the intersection form is strong enough in topological cases. Also it is known that the Kirby-Siebenmann invariant of a smoothable simply connected closed orientable 4-manifold vanishes.

In general, there are many definite quadratic form Q , i.e. $Q(\alpha, \alpha) \geq 0$ for all α or $Q(\alpha, \alpha) \leq 0$ for all α , when fixed the rank. If the rank is 32, then there are many more than 10^7 .

Theorem 1.3 (Donaldson's diagonalization theorem, 1983). *If X is a smooth closed orientable 4-manifold (originally assumed to be simply connected, but it is revealed that it is not required) with Q_X is negative-definite, then Q_X is isomorphic to $\text{diag}(-1, \dots, -1)$ over $\mathbb{Z}$.*

In particular, there should be many many non-smoothable topological 4-manifolds by the diagonalization theorem and Freedman's theorem.

High-dimensional > 4 , exotic structures can be detected by characteristic classes, which is a dominating method. Simply connected 4-manifolds, it is hard to say that there is another method to detect such phenomena...?

With boundary: Khovanov homology...

More about intersection forms:

Definition 1.4. Let $Q : L \otimes L \rightarrow \mathbb{Z}$ be a symmetric non-degenerate bilinear form over $\mathbb{Z}$, where L is a lattice, i.e. finitely generated free abelian group. Then,

$$b^+(Q) := \max \dim \text{ of positive-definite subspace of } L \otimes_{\mathbb{Z}} \mathbb{R} \text{ with respect to } Q,$$

$b^-(Q) := \max \dim$ of negative-definite subspace of $L \otimes_{\mathbb{Z}} \mathbb{R}$ with respect to Q ,

and the signature

$$\sigma(Q) := b^+(Q) - b^-(Q).$$

Note that the rank of L is $b^+(Q) + b^-(Q)$.

Also, Q is called *even* if $Q(\alpha, \alpha) \cong 0$ modulo 2 for all α , and *odd* if it not even.

When X is a closed orientable topological 4-manifold, then we use the following notations:

$$b^{\pm}(X) := b^{\pm}(Q_X), \quad \sigma(X) := \sigma(Q_X),$$

and parity and definiteness of X is defined by the parity and definiteness of Q_X .

Example 1.5.

	b^+	b^-	σ	parity
S^4	0	0	0	even
$\mathbb{CP}^2$	1	0	1	odd
$S^2 \times S^2$	1	1	0	even
$K3$	3	19	-16	even

Exercise 1.6. $Q_{K3} = 3 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \oplus 2(-E_8)$, where $-E_8$ is an even, rank 8, negative definite form, given by

$$-E_8 := \begin{pmatrix} -2 & & & & & & & \\ & -2 & 1 & & & & & \\ & & 1 & -2 & 1 & & & \\ 1 & & & 1 & -2 & 1 & & \\ & & & & 1 & -2 & 1 & \\ & & & & & 1 & -2 & 1 \\ & & & & & & 1 & -2 \end{pmatrix}.$$

See the following theorem in the course in arithmetic written by Serre.

Theorem 1.7. *Let (Q, L) be a symmetric non-degenerate bilinear form over $\mathbb{Z}$. Suppose Q is indefinite, i.e. not definite. Then, over $\mathbb{Z}$,*

$$Q \cong \begin{cases} b^+(Q)(1) \oplus b^-(Q)(-1) & \text{if } Q \text{ odd} \\ m \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \oplus n(-E_8) & \text{if } Q \text{ even and } \sigma(Q) \leq 0, \end{cases}$$

for some $m \geq 1$ and $n \geq 0$.

Other constraints on the intersection form of a smooth 4-manifold.

Theorem 1.8 (Rokhlin, 1952, before gauge theory). *Let X be a smooth closed spin 4-manifold. It is known that X admits a spin structure iff X is even, when X is simply connected. Then, 16 divides $\sigma(X)$.*

In general, 8 divides $\sigma(Q)$ if Q is even. Since $\sigma(-E_8) = -8$, so we can discover a non-smoothable manifold.

A conjecture by Y. Matsumoto, 1982: For X be a smooth closed spin 4-manifold, do we have $b_2(X) \geq \frac{11}{8} |\sigma(X)|$? Note that $b_i(X)$ is the rank of $H_i(X, \mathbb{Z})$. The equality is attained by $K3^{\#n}$ for $n \geq 1$.

Exercise 1.9. $Q_{X_1 \# X_2} = Q_{X_1} \oplus Q_{X_2}$. Use the Mayer-Vietoris.

If this 11/8-conjecture is solved, then we can obtain a complete classification of simply connected closed smooth 4-manifolds up to homeomorphism.

Exercise 1.10. If this 11/8-conjecture is solved, then simply connected closed smooth 4-manifolds are exhausted by $S^2 \times S^2$, $\mathbb{CP}^2$, $\overline{\mathbb{CP}}^2$, $K3$, $-K3$, up to homeomorphism.

There are no known examples whose smooth structures are classified.

Theorem 1.11 (Furuta, 10/8-inequality, 2001). *Let X be a smooth closed spin indefinite 4-manifold, then*

$$b_2(X) \geq \frac{10}{8}|\sigma(X)| + 2,$$

and the equality is attained by $K3$.

2 Day 2: April 21

Correction: non-degenerate $\rightarrow$ unimodular.

2.1 SW invariant

Input of SW invariant: closed oriented smooth 4-manifold X^4 with $b^+(X) \geq 2$.

Remark 2.1. There is a version for $b^+(X) = 1$. There is no reasonable definition of the SW invariant for $b^+(X) = 0$.

Output is a $\mathbb{Z}$ -valued invariant $SW(X, \mathfrak{s}) \in \mathbb{Z}$, where $\mathfrak{s}$ is a spin^c structure on X . The sign of this invariant is determined by an orientation of $H^1(X, \mathbb{R}) \oplus H^+(X)$ called a *homology orientation*. Here $H^+(X)$ is a maximal dimensional positive-definite subspace of $H^2(X, \mathbb{R})$ with respect to Q_X , meaning that $Q_X|_{H^+(X)}$ is a positive definite and $\dim H^+(X) = b^+(X)$.

Example 2.2. Let $X = \mathbb{CP}^2 \# 2\overline{\mathbb{CP}}^2$. Then, $H_2(X) = \mathbb{Z}H \oplus \mathbb{Z}E_1 \oplus \mathbb{Z}E_2 \cong \mathbb{Z}^3$, where H is hyperplane class, and E_1 and E_2 are exceptional curve classes. The smooth structure underlied on the connected sum with $\overline{\mathbb{CP}}^2$ corresponds to the blow-up at a point. In $H_2(X, \mathbb{R})$, the equation $Q_X \equiv 0$ defines a subset of a real vector space $H_2(X, \mathbb{R})$ which can be called the light cone, and $H^+(X)$ is a line in the ‘interior’ of the light cone. In this example, the set of choices of $H^+(X)$ is homeomorphic to the open unit disk.

In general, we have a following exercise, which is not easy to prove at a first glance.

Lemma 2.3. *The moduli space of linear subspaces $H \subset H^2(X, \mathbb{R})$ such that $b^+(X) = \dim H$ and $Q_X|_H$ is positive definite is contractible.*

Thanks to this lemma, if you pick one choice of $H^+(X)$, and fix an orientation on it, then it induces an orientation on another choice of $H^+(X)$ in a canonical way. Every choice of $H^+(X)$ is unique up to homotopy.

Let $\mathcal{O}$ be a homology orientation on X . Precisely, the SW invariant is an invariant of triple: $SW(X, \mathfrak{s}, \mathcal{O}) \in \mathbb{Z}$ with property that $SW(X, \mathfrak{s}, -\mathcal{O}) = -SW(X, \mathfrak{s}, \mathcal{O})$. Usually $\mathcal{O}$ is dropped from our notation.

2.2 Spin and Spin^c structures

Lemma 2.4.

$$\pi_1 \text{SO}(n) = \begin{cases} \mathbb{Z}/2\mathbb{Z} & \text{if } n \geq 3, \\ \mathbb{Z} & \text{if } n = 2. \end{cases}$$

Proof. Exercise. Use the fibration $\text{SO}(n) \rightarrow \text{SO}(n+1) \rightarrow S^n$. □

We define the spin group $\text{Spin}(n)$ as the unique double cover of $\text{SO}(n)$ for $n \geq 3$.

Definition 2.5. Let X be a oriented smooth manifold of dimension n equipped with a metric g . A *spin structure* $\mathfrak{s}$ on X is a pair (P, Ψ) , where P is a principal $\text{Spin}(n)$ -bundle over X and Ψ is an isomorphism of principal $\text{SO}(n)$ -bundles

$$\Psi : P \times_{\text{Spin}(n)} \text{SO}(n) \xrightarrow{\sim} \text{Fr}(X).$$

We should not forget Ψ , a compatibility with the data of tangent bundle. In other words, a spin structure is a choice of a lift of the frame bundle map $X \rightarrow B\text{SO}(n)$ along $B\text{Spin}(n) \rightarrow B\text{SO}(n)$. Of course it is up to homotopy because classifying spaces are defined up to homotopy equivalence, which corresponds to isomorphism classes of principal $\text{Spin}(n)$ -bundles.

Remark 2.6. To consider an isomorphism class of a spin^c structure, one does not need to pick a metric, because $\text{GL}^+(n, \mathbb{R})$ is homotopy equivalent to $\text{SO}(n)$. However, a choice of a metric is needed to define the Dirac operator.

Remark 2.7. A 4-manifold may or may not admit a spin structure. For examples, $S^2 \times S^2$ and $K3$ admit spin structures, but $\mathbb{CP}^2$ does not.

We have another exercise.

Lemma 2.8. *Let X^4 be a closed oriented smooth 4-manifold. If X admits a spin structure, then Q_X is even. The converse holds if $\pi_1(X) = 1$.*

Definition 2.9.

$$\text{Spin}^c(n) := \frac{\text{Spin}(n) \times \text{U}(1)}{\mathbb{Z}/2\mathbb{Z}},$$

where $\mathbb{Z}/2\mathbb{Z}$ acts on $\text{Spin}(n)$ as the covering transformation and on $\text{U}(1)$ faithfully as the scalar multiplication. The alphabet c stands for “complex”. We have a canonical surjection $\text{Spin}^c(n) \rightarrow \text{SO}(n)$.

Definition 2.10. Let X be a oriented smooth n -manifold equipped with a metric. A spin^c structure $\mathfrak{s}$ on X is a pair (P, Ψ) , where P is a principal $\text{Spin}^c(n)$ -bundle and Ψ is an isomorphism of principal $\text{SO}(n)$ -bundles

$$\Psi : P \times_{\text{Spin}^c(n)} \text{SO}(n) \xrightarrow{\sim} \text{Fr}(X).$$

There is a fact that every oriented 4-manifold admits a spin^c structure. (It is open that every topological 4-manifold has a CW complex structure.) We usually do not use this fact because almost every argument begin with fixing a concretely given spin^c structure. Since we have a natural $\text{Spin}(n) \rightarrow \text{Spin}^c(n)$, a spin structure induces a spin^c structure.

We have another exercise.

Lemma 2.11. *Let X be an oriented smooth manifold. If X admits a spin structure, then there is a non-canonical bijection between the set of isomorphism classes of spin structures on X and $H^1(X, \mathbb{Z}/2\mathbb{Z})$. If X admits a spin^c structure, then there is a non-canonical bijection between the set of isomorphism classes of spin^c structures on X and $H^2(X, \mathbb{Z})$.*

If $\pi_1(X) = 1$, then spin structure is unique up to isomorphism.

orientability iff $w_1 = 0$, spin iff $w_2 = 0$, spin^c iff w_2 is lifted to $H^2(X, \mathbb{Z})$.

2.3 SW equations

Let X be an oriented smooth 4-manifold with a metric g and a spin^c structure $\mathfrak{s}$. We can write down the SW equation

$$\begin{cases} F_A^+ = \sigma(\Phi) \\ D_A \Phi = 0 \end{cases}$$

where (A, Φ) is unknown functions. Here A is a $\text{U}(1)$ -connection on the determinant line bundle for $\mathfrak{s}$, and Φ is a positive spinor for $\mathfrak{s}$. Roughly, $\text{SW}(X, \mathfrak{s})$ is the signed count of

$$\frac{\#\{\text{solutions } (A, \Phi)\}}{\text{gauge symmetry}}.$$

It is typically an oriented compact 0-dimensional manifold.

$b^+ \geq 2$, the singularity with codimension 1?

Lemma 2.12.

$$\mathrm{Spin}(4) \cong \mathrm{Sp}(1) \times \mathrm{Sp}(1), \quad \mathrm{Sp}(1) = S(\mathbb{H}).$$

Proof. $\mathrm{Sp}(1) \times \mathrm{Sp}(1) \rightarrow \mathrm{SO}(\mathbb{H}) : (q_+, q_-) \mapsto (q \mapsto q_- q_+)$ is a double cover. □

Since $\mathrm{Sp}(1) \cong \mathrm{SU}(2) : a + jb \mapsto \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}$ canonically, we have four representations of

$$\mathrm{Spin}^c(4) = \frac{\mathrm{SU}(2) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}/2\mathbb{Z}}$$

given by

$$\Delta_+ : \mathrm{Spin}^c(4) \rightarrow \mathrm{U}(2) : [q_+, q_-, \lambda] \mapsto \lambda q_+,$$

$$\Delta_- : \mathrm{Spin}^c(4) \rightarrow \mathrm{U}(2) : [q_+, q_-, \lambda] \mapsto \lambda q_-,$$

$$\det : \mathrm{Spin}^c(4) \rightarrow \mathrm{U}(1) : [q_+, q_-, \lambda] \mapsto \lambda^2.$$

These representations induces positive spinor bundle S^+ , negative spinor bundle S^- , and the determinant line bundle L .

We will frequently use the multiplication of quaternions for computations.

3

What is Ω^+ ? it is related to $b^+ > 0$?

4 Day 4: May 12

Let X^4 be an oriented closed smooth 4-manifold together with a metric g and a spin^c -structure $\mathfrak{s}$. The configuration space of the SW equation is

$$\mathcal{C}(X, \mathfrak{s}, g) := \mathcal{A}(L) \times \Gamma(S^+).$$

The gauge group is

$$\mathcal{G}(X) := C^\infty(X, \text{U}(1)).$$

Then, $\mathcal{G}$ acts from right(?) on $A \in \mathcal{A}(L)$, $\Phi \in \Gamma(S^\pm)$, $\omega \in i\Omega^+(X)$ such that

$$A \cdot u := u^* A := u^{-1} du + A, \quad \Phi \cdot u := \Phi u, \quad \omega \cdot u := \omega.$$

Let $\eta \in i\Omega^+$. The η -perturbed SW equations are

$$\begin{cases} F_A^+ = \sigma(\Phi) + \eta, \\ D_A \Phi = 0. \end{cases}$$

It is introduced to get a degree of freedom for transversality.

Lemma 4.1. *Let $\eta \in i\Omega^+$. Then, $\text{SW}_\eta : \mathcal{C} \rightarrow i\Omega^+ \oplus \Gamma(S^-) : (A, \Phi) \mapsto (F_A^+ - \sigma(\Phi) - \eta, D_A \Phi)$ is $\mathcal{G}$ -equivariant.*

Proof. Let $u \in \mathcal{G}$.

First,

$$F_{u^* A}^+ = F_{u^{-1} du + A}^+ = p^+ d(u^{-1} du + A) = p^+ d(u^{-1} du) + F_A^+.$$

Here, lifting $u : U \subset X \rightarrow \text{U}(1)$ to get $\tilde{u} : U \subset X \rightarrow \mathbb{R}$ locally in the sense that $u = e^{2\pi i \tilde{u}}$, it is locally exact, so it is globally closed

$$d(u^{-1} du) = d((2\pi i)^{-1} d \log u) = 0,$$

and $F_{u^* A}^+ = F_A^+$.

Second, recall that $\sigma(\Phi)$ is defined by a $\text{Spin}^c(4)$ -equivariant map $\sigma : \mathbb{H}_{S^+} \rightarrow \text{Im } \mathbb{H}_{\Lambda^+} : \Phi \mapsto \Phi i \bar{\Phi}$, so it is clear.

Third, recall that $\nabla_A : \Gamma(S^+) \rightarrow \Gamma(T^*X \otimes S^+)$ is $\mathcal{G}$ -equivariant by $\nabla_{u^* A}(\Phi u) = (\nabla_A \Phi) \cdot u$. On the Clifford multiplication, it is $c : T^*X \otimes S^+ \rightarrow S^-$ induced from the quaternion multiplication $\mathbb{H}_{T^*X} \otimes \mathbb{H}_{S^+} \rightarrow \mathbb{H}_{S^-}$, so it is $\mathcal{G}$ -equivariant. So we are done. $\square$

By the lemma, the solution space $\text{SW}_\eta^{-1}(0)$ has a right $\mathcal{G}$ -action. We define the *moduli space of solutions to the (η -perturbed) SW equations* is

$$\mathcal{M}(X, \mathfrak{s}, g, \eta) := \text{SW}_\eta^{-1}(0) / \mathcal{G}.$$

It may not be good to call this the “moduli space”, because it has no relation with stable bundles, representable functors, etc.

Lemma 4.2. *If $b^+ > 0$, for “generic” (g, η) , $\text{SW}_\eta^{-1}(0)$ is a (infinite-dimensional) submanifold of $\mathcal{C} = \mathcal{A}(L) \times \Gamma(S^+) \hookrightarrow i\Omega^1(X) \oplus \Gamma(S^+)$.*

Sketch. For the space $\text{Met}(X)$ of all metrics on X , on the space $\bigcup_{g \in \text{Met}(X)} \Omega_g^+$, we can consider an affine

bundle $\bigcup_{g \in \text{Met}(X)} (\mathcal{C}_g \times \Omega_g^+)$. Then, we can prove

$$\bigcup_{g \in \text{Met}(X)} \bigcup_{\eta \in i\Omega_g^+} \text{SW}_\eta^{-1}(0) \subset \bigcup_{g \in \text{Met}(X)} (\mathcal{C}_g \times \Omega_g^+)$$

is a submanifold.

Then, we finally get the desired consequence by the implicit function theorem and the Sard-Smale theorem for infinite-dimensional manifolds. $\square$

Theorem 4.3. *If $b^+ > 0$, for “generic” (g, η) , $\mathcal{M}$ is a smooth **closed** manifold of dimension*

$$d(\mathfrak{s}) := \frac{1}{4}(c_1(\mathfrak{s})^2 - 2\chi(X) - 3\sigma(X)) \in \mathbb{Z},$$

where $c_1(\mathfrak{s}) := c_1(L)$, $c_1(\mathfrak{s})^2 = \langle c_1(\mathfrak{s})^2, [X] \rangle \in \mathbb{Z}$. It is a consequence of the Atiyah-Singer index theorem and the Hausdorffness of $\mathcal{G}(?)$.

The number $d(\mathfrak{s})$ is called the *formal dimension*, and it maybe negative, and in this case $\mathcal{M} = \emptyset$. The compactness of $\mathcal{M}$ is the most important property of SW theory. An orientation, also called the homology orientation, on $H^1(x, \mathbb{R}) \oplus H^+(X)$ gives rise to an orientation of $\mathcal{M}$. Define

$$\text{SW}(X, \mathfrak{s}) := \begin{cases} \#\mathcal{M} & \text{if } d(\mathfrak{s}) = 0, \\ \langle \mu^d, [\mathcal{M}] \rangle & \text{if } d(\mathfrak{s}) \in 2\mathbb{Z}_{\geq 0}, \\ 0 & \text{if } d(\mathfrak{s}) \text{ is odd or negative,} \end{cases}$$

where $\mu \in H^2(\mathcal{B}, \mathbb{Z})$ is defined as follows: We define $\mathcal{B}^* := \mathcal{C}^*/\mathcal{G} \supset \mathcal{M}$. Pick $x_0 \in X$ and let $\mathcal{G}_0 := \{u \in \mathcal{G} : u(x_0) = 1\}$. Since $U(1)$ freely acts on $\tilde{\mathcal{B}}^* := \mathcal{C}^*/\mathcal{G}_0$, we can define $\mu := c_1(\tilde{\mathcal{B}} \rightarrow \mathcal{B})$.

Theorem 4.4. *If $b^+ \geq 2$, $\text{SW}(X, \mathfrak{s})$ is independent of the choice of generic (g, η) .*

Remark 4.5. If $b^+ = 1$, we get a pair $(\text{SW}^+(X, \mathfrak{s}), \text{SW}^-(X, \mathfrak{s})) \in \mathbb{Z}^2$ as an invariant of $(X, \mathfrak{s})$. For $b_1 = 0$, wall-crossing formula $\text{SW}^+(X, \mathfrak{s}) = \text{SW}^-(X, \mathfrak{s}) \pm 1$

Remark 4.6 (Simple type conjecture). If $b^+ \geq 2$, then $\text{SW}(X, \mathfrak{s}) = 0$ if $d(\mathfrak{s}) > 0$.

So $d(\mathfrak{s}) = 0$ is important for computations.

4.1 How to think of the formal dimension

An equivariant map $\text{SW}_\eta : \mathcal{C} \rightarrow i\Omega^+ \oplus \Gamma(S^-) =: V$ induces a section of $s : \mathcal{C} \times_{\mathcal{G}} V =: \mathcal{E} \rightarrow \mathcal{B} := \mathcal{C}/\mathcal{G}$. Then, $\mathcal{M} := s^{-1}(0)$, and intuitively $d(\mathfrak{s}) = \dim \mathcal{B} - \dim V$. It is rigorously given by the Fredholm index of $ds : T_b \mathcal{B} \rightarrow V$ at $b \in \mathcal{M} \subset \mathcal{B}$, and computed by the Atiyah-Singer index.

Lemma 4.7. *Let $(A, \Phi) \in \mathcal{C}$. Then, the stabilizer of the action by $\mathcal{G}$ on $\mathcal{C}$ at (A, Φ) is*

$$\text{Stab}_{(A, \Phi)} = \begin{cases} \{1\} & \text{if } \Phi \neq 0, \\ U(1) & \text{if } \Phi = 0. \end{cases}$$

Proof. If $\Phi \neq 0$, for $u \in \mathcal{G}$, the stabilization means $du = 0$ with $\Phi(x_0)u(x_0) = \Phi(x_0)$ for some x_0 with $\Phi(x_0) \neq 0$.

If $\Phi = 0$, then $(A, 0)u = (A, 0)$ implies $u = 1$. $\square$

Therefore, $\mathcal{G}$ acts from right on $\mathcal{C}^* := \mathcal{C} \setminus (\mathcal{A}(L) \times \{0\})$.

5 Day 5: May 26

X^4 is an oriented

Definition. $(A, \Phi) \in \mathcal{C}$ is called *irreducible* if $\Phi \neq 0$ in $\Gamma(S^+)$.

It comes from the relation to algebraic geometry and the decomposability of vector bundles in Donaldson theory.

We want to know when the perturbed SW equations have a reducible solution. If we let the space of perturbations $\Pi(X) := \bigcup_{g \in \text{Met}(X)} i\Omega_g^+$, then $\Pi(X) \rightarrow \text{Met}(X)$ forms a Hilbert bundle. For $g \in \text{Met}(X)$,

$$\mathcal{W}_g(X) := F_{A_0}^+ + \text{Im}(d^+ : i\Omega^1 \rightarrow i\Omega_g^+) \subset \Pi_g(X).$$

Then, $\mathcal{W}_g(X)$ does not depend on the choice of A_0 because $A_0, A_1 \in \mathcal{A}(L)$ has $F_{A_1}^+ - F_{A_0}^+ = d^+(A_1 - A_0) \in \text{Im } d^+$. Now we let $\mathcal{W}(X) := \bigcup_{g \in \text{Met}(X)} \mathcal{W}_g(X)$.

Lemma. For $\eta \in i\Omega_g^+$, the η -perturbed SW equations have a reducible solution iff $\eta \in \mathcal{W}_g(X)$.

Proof. Suppose there is a reducible solution. It means that $\Phi = 0$, so $F_A^+ = \eta$.

Conversely, if $\eta \in \mathcal{W}_g(X)$, then there is $a \in i\Omega^1$ such that $\eta = F_{A_0}^+ + d^+a = F_{A_0+a}^+$. Thus we have a reducible solution. $\square$

Proposition. The complement $\Pi(X) \setminus \mathcal{W}(X)$ is homotopy equivalent to $S^{b^+(X)-1}$.

Proof. $\mathcal{W}_g(X) = F_{A_0}^+ + \text{im } d^+$ is an affine subspace of $i\Omega^+$ of codimension $\dim \mathcal{H}^+ = b^+$ by the Hodge theory. Then, since $\text{Met}(X)$ is contractible, we are done. $\square$

Remark (Hodge theory). In general, let (X, g) be an oriented closed Riemannian manifold. For $d^* := (-1)^k *^{-1} d^* : \Omega^k \rightarrow \Omega^{k-1}$, it is easy to show $(d\omega, \eta)_{L^2} = (\omega, d^*\eta)_{L^2}$, where $\omega \in \Omega^k$, $\eta \in \Omega^{k-1}$, and $(\alpha, \beta) := \int \alpha \wedge * \beta$ for $\alpha, \beta \in \Omega^k$. The Laplacian is defined by $dd^* + d^*d$ which acts on Ω^k . Let $\mathcal{H}^k(X) := \ker \Delta \subset \Omega^k$. The Hodge theorem states that

$$\Omega^k = \mathcal{H}^k \oplus \text{im}(d : \Omega^{k-1} \rightarrow \Omega^k) \oplus \text{im}(d^* : \Omega^{k+1} \rightarrow \Omega^k).$$

For $\dim X = 4$, set $\mathcal{H}^+ := \mathcal{H}^2 \cap \Omega^+$, called harmonic self-dual 2-forms.

We prove that $\dim \mathcal{H}^+(X) = b^+(X)$, $\dim \mathcal{H}^-(X) = b^-(X)$. Before the proof, recall that there is also a canonical isomorphism $\mathcal{H}^k \rightarrow H_{\text{dR}}^k$ once fixing a metric. Since $Q_X|_{\mathcal{H}^\pm(X)}$ is positively/negatively definite, $\mathcal{H}^\pm(X)$ gives a model of $H^\pm(X)$. Now give the proof. For $\omega \in \mathcal{H}^+(X)$,

$$Q_X([\omega], [\omega]) = \int_X \omega \wedge \omega = \int_X \omega \wedge * \omega = \|\omega\|_{L^2}^2 \geq 0.$$

Similarly, $\omega \in \mathcal{H}^-(X)$ has

$$Q_X([\omega], [\omega]) = \int_X \omega \wedge \omega = - \int_X \omega \wedge * \omega = -\|\omega\|_{L^2}^2 \leq 0.$$

So we are done.

We also prove that $\Omega^+ = \mathcal{H}^+ \oplus \text{im}(d^+ : \Omega^1 \rightarrow \Omega^+)$. If $h \in \mathcal{H}^*$, then $*h \in \mathcal{H}^{\dim X - *}$. In the decomposition $\Omega^2 = \mathcal{H}^2 \oplus \text{im } d \oplus \text{im } d^*$, if $\omega = h + d\alpha + d^*\beta$, then $*\omega = *h + *d^*\beta + *d\alpha$, so $h = *\omega$. Thus, $\omega \in \Omega^+$ has

$$\omega = h + d\alpha + d^*\beta = h + d\alpha + *d\alpha = h + d^+(2\alpha) \in \mathcal{H}^+ \oplus \text{im } d^+.$$

Further, $d^+(2\alpha) = d\alpha + d^*\beta \in \text{im } d \oplus \text{im } d^*$, so $\Omega^+ = \mathcal{H} \oplus \text{im } d^+$.

Proposition. *If $b^+ \geq 2$, then for “generic” $\eta_0, \eta_1 \in \Pi(X)$, $\#\mathcal{M}(X, \mathfrak{s}, g_0, \eta_0) = \#\mathcal{M}(X, \mathfrak{s}, g_1, \eta_1)$, where g_0 and g_1 are base-points of η_0 and η_1 .*

Proof. For simplicity assume $d(\mathfrak{s}) = 0$. Since $\Pi(X) \setminus \mathcal{W}(X)$ is path-connected, there is a “generic” path η_t such that $\bigcup_{t \in [0,1]} \mathcal{M}(\eta_t)$ is a compact oriented 1-manifold, whose the boundary has zero signed count. (Such an argument is called the “cobordism argument”.) $\square$

When $b^+ = 1$, an orientation on $H^+(X)$ determines which component of $\Pi(X) \setminus \mathcal{W}(X) \simeq S^0$ is “positive”. It defines SW^+ and SW^- for each component, which is independent of (g, η) .

Analytic preliminaries

A differential operator $D : \Gamma(E) \rightarrow \Gamma(F)$ of order k gives rise to a map $\sigma(D) : T^*X \rightarrow \text{Hom}(E, F)$ such that

$$\sum_{i=1}^d u_i dx_i \mapsto \sum_{|\alpha|=k} a_\alpha(x) u^\alpha,$$

where $u^\alpha := u_1^{\alpha_1} \cdots u_d^{\alpha_d}$. Then, $\sigma(D)$ is independent of charts and local trivializations. It is not $C^\infty(M)$ -linear unless $k = 1$. If $E \cong F$ and $\sigma(D) : T^*X_0 \rightarrow \text{Hom}(E, F)_0$, then we call D an *elliptic operator*.

6

elliptic complexes

1. de Rham complex
2. Atiyah-Hitchin-Singer complex

$$0 \rightarrow \Omega^0 \rightarrow \Omega^1 \rightarrow \Omega^+ \rightarrow 0$$

3. spin^c 4-manifold,

$$0 \rightarrow \Gamma(S^+) \xrightarrow{D_A} \Gamma(S^-) \rightarrow 0$$

Let X be a closed oriented manifold, and E is a vector bundle. Fix a metric on X and E , and fix a connection on E .

$$\begin{aligned} \|s\|_{W^{k,p}(E)} &:= \left(\sum_{i=0}^k \int_X |\tilde{\nabla}^{(i)} s|^p d\text{vol} \right)^{1/p}, \\ \tilde{\nabla}^{(i)} &= \nabla^{(i-1)} \circ \dots \circ \nabla^{(0)} : \Gamma(E) \rightarrow \Gamma(T^*X^{\otimes i} \otimes E), \\ \nabla^{(i)} &:= \text{id}_{T^*X^{\otimes i}} \otimes \nabla + \nabla_{\text{LC}} \otimes \text{id}_E. \end{aligned}$$

6.1

Let X be a spin^c closed smooth 4-manifold equipped with a metric g .

$$\text{SW} : \mathcal{A}(L) \times \Gamma(S^+) \rightarrow i\Omega^+ \oplus \Gamma(S^-)$$

is (non-trivially) extended to

$$\text{SW} : H^k(i\Lambda^1) \otimes H^k(S^+) \rightarrow H^{k-1}(i\Lambda^+) \oplus H^{k-1}(S^-)$$

for $k \geq 3$.

This follows from the boundedness of the multiplication $H^k \otimes H^k \rightarrow H^{k-1}$ for $k \geq 3$ because on $d = 4$ we have Sobolev embeddings $H^3 \subset C^0$ and $H^2 \subset L^4$.

If

$$d\text{SW} : T\mathcal{C}_{H^k} := T(\mathcal{A}_{H^k} \times H^k(S^+)) \rightarrow H^{k-1}(i\Lambda^+) \oplus H^{k-1}(S^-)$$

surjective at every point, then the implicit function theorem says that $\text{SW}^{-1}(0)$ is a Hilbert manifold.

The gauge group $\mathcal{G}_{H^{k+1}} := H^{k+1}(X, \mathbb{C}) \cap C^0(X, \text{U}(1))$ is a Hilbert Lie group. It acts on the domain and the codomain of the Seiberg-Witten map smoothly. Then, $\mathcal{M} := \text{SW}^{-1}(0)/\mathcal{G}_{H^{k+1}}$ is a smooth manifold. Following is the proof idea.

$\mathcal{G}_{H^{k+1}}$ -action has local slices on $\mathcal{C}_{H^k}$, i.e. for every $\gamma \in \mathcal{C}_{H^k}^*$, there is a submanifold $S \subset \mathcal{C}_{H^k}^*$ containing γ such that $\mathcal{G}_{H^{k+1}} \times S \rightarrow \mathcal{C}_{H^k}^*$ gives a diffeomorphism onto a neighborhood of the orbit $\mathcal{G}_{H^{k+1}} \cdot \gamma$. This slice S gives a local chart of $\mathcal{B}_{H^k}^* := \mathcal{C}_{H^k}^*/\mathcal{G}_{H^{k+1}}$, making $\mathcal{B}_{H^k}^*$ a Hilbert manifold. $\text{SW}^{-1}(0) \cap S$ gives a local slice for the action of $\mathcal{G}$ on $\text{SW}^{-1}(0)$, giving a local chart for $\mathcal{M}$ near $[\gamma]$.

In fact, Hausdorff is not trivial. We need to show $\mathcal{B}_{H^k}^*$ is Hausdorff. This follows from the compactness of $\text{U}(1)$.

The dimension of $\mathcal{M}$ is computed by the middle cohomology of

$$T_e \mathcal{G}_{H^{k+1}} \rightarrow T_\gamma \mathcal{C}_{H^k} \rightarrow H^{k-1}(i\Lambda^+) \oplus H^{k-1}(S^-).$$

Ignoring zeroth order terms, this is the direct sum of the Atiyah-Hitchin-Singer complex and the Dirac operator complex $0 \rightarrow 0 \rightarrow \Gamma(S^+) \rightarrow \Gamma(S^-) \rightarrow 0$, which is eventually the index, computed by the Atiyah-Singer to be $(-1 + b_1(X) - b^+(X)) + \frac{1}{4}(c_1(\mathfrak{s})^2 - \sigma(X)) = \frac{1}{4}(c_1(\mathfrak{s})^2 - 2\chi(X) - 3\sigma(X))$.